



EDUCATION AND SCHOLASTIC ACHIEVEMENTS

Program	Institution	CGPA/%	Year
BTech in Biological Engineering & MTech in Quantitative Finance	Indian Institute of Technology Madras	8.51/10	2027
Class XII	Baldeo Public School, Baldeo, Mathura	93.16%	2021
Class X	Baldeo Public School, Baldeo, Mathura	95.60%	2019

COURSES AND SKILLS

Courses	MS6621 Corporate Finance	HS3021 Macroeconomics
	HS3002 Principles of Economics	MA5755 Data Analysis & Visualization in R/Python/SQL
	BT3041 Analysis and Interpretation of Biological Data	MA5890 Numerical Linear Algebra
	MA5950 Mathematical Finance**	MA2040 Probability, Statistics and Stochastic Processes
Skills	C++, Python, Matlab, NumPy, Pandas, STL, SQL, Tableau, MS Office, Latex	

TECHNICAL EXPERIENCE/PROJECTS

Black Scholes Option Pricing Calculator Project	- Developed Black-Scholes option pricing engine in Python using closed-form solutions & continuous compounding - Derived analytical expressions for key Greeks to enable sensitivity analysis across volatility, strike price, and time. - Plotted option prices and Greek surfaces across multiple dimensions to visualize market parameter relationships.
Gene - Heart Failure Risk Analysis Academic Project BT3041	- Designed an end-to-end machine learning pipeline for causative gene selection and early heart failure risk prediction. - Carried comprehensive EDA² & dimensionality reduction using PCA & clustering to assess gene expression patterns. - Used variance thresholding & K-best selection; ranked influential genes using Gini index on trained RF³ classifiers. - Built a logistic regression model yielding 91% accuracy, narrowly surpassing the original study's reported baseline.
Financial Ratios Academic Project MS6621	- Analysed Apollo Hospitals Enterprises Limited's financials by evaluating 5+ fundamental financial parameters. - Calculated 30+ financial ratios by thoroughly studying profitability, liquidity, solvency, turnover, & valuation metrics. - Benchmarked the financial ratios against market standards getting insights for assessing long-term investment risks.
Portfolio Optimization Project	- Optimized Nifty Commodities Index portfolio, successfully reducing the risk by 20% while maintaining the returns. - Constructed minimum variance and Markowitz portfolios and significantly improved the Sharpe Ratio by 75.2%. - Created detailed and insightful visualizations of the Efficient Frontier and Indifference Curve using plotly library.

VIRTUAL EXPERIENCE⁴

J.P. Morgan Chase and Co. Quantitative Research	- Built a robust pricing model to estimate natural gas prices by SARIMA⁵ forecasting model on historical market data. - Developed a python function to calculate price storage contracts, factoring in dates, prices, rates, volume, and costs. - Implemented quantization algorithm using dynamic programming to map FICO⁶ scores to the customer ratings. - Built a Random Forest model to predict probability of loan default and expected loss with an accuracy of over 90%. - Optimized credit rating buckets by maximizing log-likelihood function, significantly improving the default estimates.
Citi Market Sales & Trading	- Monitored hedge funds and pension flows; analyzed yield curve shifts following FOMC⁷ updates & rate projections. - Pointed mispricing in FOMC OIS⁸; proposed trade anticipating persistent inflation & earlier-than-expected Fed hikes. - Developed UST⁹ and IRS¹⁰ trade strategies; supported by macroeconomic data, yield curve analysis, and Fed outlook. - Used interest rate swaps, FOMC-linked trades, and Treasury futures to express curve, duration, and policy views.

POSITIONS OF RESPONSIBILITY

Co-founder and Head, Plant Based Society, IITM	- Co-founded a new club to raise awareness & promote animal welfare, rights and plant-based lifestyles on the campus. Raised INR 100K+ in funding for logistical operations from 2 non-profit organisations & numerous helpful alumni. - Interacted with over 1000 people via 15+ events, including outreaches, speaker talks, screenings, and campaigns, etc. - Hosted various prominent speakers like Shri Acharya Prashant, Dr. Sailesh Rao, & Arvind Kannan for discussions.
IITM Team Head, Shiksha Prayas, NGO	- Spearheaded a team of 20 first year students for mentoring 40+ underprivileged students from the outskirts of Delhi. - Conducted online presentations on career opportunities for more than 100 students in 4 centers of the organisation. - Developed basic computer programming curriculum for sub-secondary students in Scratch programming language.
Research Analyst, The Fifth Estate, IITM	- Developed and conducted a pan-IIT survey assessing student perception and effectiveness of governance across IITs. - Curated and conducted 2025 placement survey seeking insights on various parameters affecting students' placements. - Made a comprehensive review of outgoing Academic Affairs Secretary's election manifesto and overall performance.

EXTRA-CURRICULAR ACTIVITIES

- Achieved selection in prestigious **NDA¹¹** and invited to undergo the rigorous 5-day SSB¹² interviews at both **SSB Bhopal**, and **SSB Allahabad**.
- Volunteered in numerous freshie mentorship programs by being a **Saathi Mentor**, **AMC¹³ Mentor**, and **FacAd Mentor** for over **15 students**.
- Managing a **writing blog** on philosophical commentaries, ethics, animal ethics, culture, and modern social dynamics with over **20 entries** live.

** , Upcoming course. 2, Exploratory Data Analysis. 3, Random Forests. 4, Forage Work Simulation 5, Seasonal Autoregressive Integrated Moving Average. 6, Fair Isaac Corporation Score. 7, Federal Open Market Committee. 8, Overnight Index Swap. 9, United States Treasury. 10, Interest Rate Swaps. 11, National Defence Academy. 12, Service Selection Board. 13, Academic Mentorship Cell.